

Three-dimensional radiation directivity of six violins built with distinct making techniques: measurement, spherical-harmonic decomposition and comparative analysis

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The three-dimensional far-field radiation directivity of six violins, manufactured with identical external dimensions but different lutherie techniques, is measured and compared in the frequency range 10 Hz to 10 000 Hz. Each violin is positioned inside a parallelepipedal microphone array of 256 sensors; impact hammer excitation is applied five times at the top of the bridge with strings mounted in normal playing condition. From the 256 pressure-over-force frequency response functions (FRFs), the Helmholtz Equation Least-Squares (HELs) method identifies the complex spherical-harmonic (SH) expansion coefficients up to order $N = 7$ (64 coefficients), yielding the complex far-field directivity $D(\theta, \varphi, \omega)$ and its time-domain counterpart $d(\theta, \varphi, t)$ via inverse Fourier transform. A comprehensive set of scalar and vectorial indices — spanning modal energy content, active directivity vector, phase coherence, group delay, impulse-energy maps and reduced-space comparison — is derived and interpreted in terms of radiation physics and lutherie practice. Significant differences among violins are evidenced both in directivity amplitude and, crucially, in directivity phase, which encodes causal and modal coupling information inaccessible to amplitude-only analyses.

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I. INTRODUCTION

The violin has been the subject of acoustic inquiry for over a century, yet the quantitative link between lutherie practice and the acoustic behaviour of the finished instrument remains incompletely understood. The Bilbao Project addresses this gap by combining controlled manufacturing protocols — identical external geometry, variable internal structure and wood selection — with systematic acoustic characterisation.

Among the acoustic quantities that bear directly on the perceived quality of a string instrument, the far-field radiation directivity occupies a central position: it determines how sound energy is distributed in angle as a function of frequency and therefore shapes both the in-room sound field and the auditory impression of listeners at different positions. Early fundamental work by , and later established that directivity is intimately related to the vibration modes of the body and to the coupling between structural and acoustic resonances. Amplitude-based directivity “balloon” representations at selected frequencies are now well established in the literature (;), yet the *phase* of the directivity function — which encodes causal relationships, inter-modal coupling and the

direction-dependent group delay — has received comparatively little attention.

Spherical-harmonic (SH) decomposition (;) provides a natural framework for compact, physically meaningful descriptions of three-dimensional radiation fields. The HELs (Helmholtz Equation Least Squares) method (;) allows the reconstruction of the SH expansion from a finite, non-conformal microphone array, making it compatible with practical laboratory setups. This work leverages both tools to build a rich comparative portrait of six violins across the full musical frequency range.

The remainder of the paper is organised as follows. Section II describes the experimental setup and the six violins. Section III details the signal-processing pipeline from raw FRFs to complex far-field directivity. Sections IV–IX present the results in order of increasing abstraction: FRF validation, SH coefficient analysis, frequency-domain directivity, time-frequency analysis, impulse directivity and multi-violin comparison. Section X interprets the findings in terms of lutherie technique. Section XI concludes.

II. EXPERIMENTAL SETUP

A. The six violins of the Bilbao Project

Six full-size violins were manufactured specifically for the project. External dimensions, including overall body length, upper/lower bout widths and arching heights, are identical across instruments, following the specifications described in [Bilbao Project reference]. The six instruments differ in [description of lutherie variables: wood species, tap-tone selection, varnish type, graduation thicknesses, bass-bar dimensions, etc. — to be completed by authors]. All violins were fitted with the same brand and gauge of strings and adjusted with identical sound-post positions and bridge geometry to isolate the effect of the lutherie variables.

B. Microphone array and geometry

The array is a parallelepipedal cage of dimensions $[L_x \times L_y \times L_z]$ m carrying 256 microphones (model [to specify]) arranged on [to specify: regular grid / irregular distribution]. The acoustic centre is fixed at the midpoint of the top face of the bridge (hammer impact point), which serves as the origin of the SH expansion. The array geometry, microphone positions and their distances from the acoustic centre are provided as supplementary material.

C. Impact hammer excitation and signal acquisition

Five impacts are applied at the top of the bridge with an instrumented impact hammer (model [to specify], tip [to specify]). Strings are mounted in normal playing condition (tension, mute absent), so that the instrument responds as it would in actual use, including the mechanical loading of the tailpiece–string system. The excitation force $F(t)$ and acoustic pressures $p_k(t)$ ($k = 1, \dots, 256$) are recorded simultaneously at a sampling frequency of 50 kHz over a window of 1 s per impact. A pre-trigger of 1 ms is included for causality verification (see Section IV B).

III. SIGNAL PROCESSING AND HELS RECONSTRUCTION

A. FRF estimation

The $H_k(\omega) = P_k(\omega)/F(\omega)$ FRF for each microphone k is estimated by the H_1 estimator averaged over the five impacts:

$$H_k(\omega) = \frac{\sum_{l=1}^5 P_k^{(l)}(\omega) \overline{F^{(l)}(\omega)}}{\sum_{l=1}^5 |F^{(l)}(\omega)|^2}, \quad (1)$$

where the overbar denotes complex conjugation. The ordinary coherence function

$$\gamma_k^2(\omega) = \frac{|\langle P_k F^* \rangle|^2}{\langle |P_k|^2 \rangle \langle |F|^2 \rangle} \quad (2)$$

is used to validate the linearity and signal-to-noise quality of each FRF.

B. HELS reconstruction

The HELS method (;) reconstructs the acoustic pressure field in the vicinity of the source from the measured FRFs by projecting them onto outgoing spherical wave solutions of the Helmholtz equation. The pressure in the far field is expressed as

$$p(r, \theta, \varphi, \omega) \xrightarrow{kr \gg 1} \frac{e^{jkr}}{r} \sum_{n=0}^N \sum_{m=-n}^n a_n^m(\omega) Y_n^m(\theta, \varphi), \quad (3)$$

where $Y_n^m(\theta, \varphi)$ are the real-valued spherical harmonics, $k = \omega/c$ is the wavenumber, and the complex coefficients $a_n^m(\omega)$ are determined by a regularised least-squares inversion of the measurement equations at the 256 array positions. The expansion is truncated at order $N = 7$ (64 independent coefficients), consistent with the spatial Nyquist limit imposed by the inter-microphone spacing and the maximum frequency of interest ().

The complex far-field directivity pattern is then defined as

$$D(\theta, \varphi, \omega) = \sum_{n=0}^N \sum_{m=-n}^n a_n^m(\omega) Y_n^m(\theta, \varphi), \quad (4)$$

and the impulse directivity is its inverse Fourier transform:

$$d(\theta, \varphi, t) = \mathcal{F}^{-1}\{D(\theta, \varphi, \omega)\}. \quad (5)$$

IV. FRF VALIDATION AND CAUSALITY

A. Coherence maps

What it shows: The coherence $\gamma_k^2(\omega)$ quantifies the linear, noise-free fraction of the FRF at each microphone and frequency. Values close to unity confirm that the measurement is not corrupted by background noise, nonlinearities or multiple incoherent excitation paths. Regions of low coherence (typically at anti-resonances or below the cut-on frequency of the source) indicate where FRF estimates carry larger uncertainty.

Representation: A 2D map of γ_k^2 as a function of microphone index and frequency, supplemented by a frequency-averaged coherence curve $\bar{\gamma}^2(\omega)$. Threshold lines at $\gamma^2 = 0.9$ and $\gamma^2 = 0.95$ are superimposed.

Comparative value: Consistent high coherence across all six violins validates the measurement protocol and confirms that differences in FRFs originate from the instruments rather than experimental artefacts.

References: , .

B. Causality check

What it shows: The impulse response $h_k(t) = \mathcal{F}^{-1}\{H_k\}$ must be causal: no energy should appear before the sound travel time $t_{min} = d_{min}/c$ from source

to microphone. A non-causal impulse response indicates phase errors in the FRF (e.g., windowing artefacts, clock drift). The excess phase $\phi_{exc}(\omega) = \angle H_k - \angle H_k^{min}$ (difference between measured and minimum-phase counterpart) provides additional information on the all-pass component of the radiation.

Representation: Waveform plots of $h_k(t)$ for representative microphones with the causal onset marked; histogram of excess-phase norms across the 256 channels.

References: , .

V. ANALYSIS OF SPHERICAL-HARMONIC COEFFICIENTS

A. Modal energy spectra

The energy carried by SH order n as a function of frequency is

$$E_n(\omega) = \sum_{m=-n}^n |a_n^m(\omega)|^2. \quad (6)$$

The total radiated power (up to a constant factor) is

$$W(\omega) = \sum_{n=0}^N E_n(\omega). \quad (7)$$

What it shows: $E_n(\omega)$ reveals at which frequencies the radiation becomes spatially complex (high-order modes activated) vs. quasi-monopolar or dipolar (low orders dominate). The transition from low-order to high-order dominated radiation is a signature of the modal density of the body.

Representation: (a) 2D colour maps of E_n in the $(n, \log f)$ plane for each violin; (b) $E_n(f)$ curves superimposed for all six violins, one panel per order n .

Comparative value: Differences in E_n across violins at the same frequency identify instruments in which specific vibrational modes radiate more or less efficiently in each angular complexity class.

References: (Ch. 4), , (Ch. 3).

B. Effective modal order and Shannon entropy

$$N_{eff}(\omega) = \exp\left(-\sum_{n=0}^N \tilde{E}_n(\omega) \ln \tilde{E}_n(\omega)\right), \quad \tilde{E}_n = \frac{E_n}{W}, \quad (8)$$

where $\tilde{E}_n$ are normalised modal energies.

What it shows: N_{eff} is a scalar index of the spatial complexity of the radiation: $N_{eff} = 1$ corresponds to pure monopole radiation; $N_{eff} = N + 1$ to perfectly equipartitioned radiation. High values at specific frequencies indicate multimodal, complex directivity patterns.

Representation: $N_{eff}(f)$ curves for the six violins, averaged in one-third-octave bands. Shaded regions highlight modes A0, B1⁻, B1⁺ identified on the power spectrum.

Comparative value: A violin with systematically lower N_{eff} radiates in a more “focused” way (fewer active modes); higher N_{eff} indicates more diffuse, omnidirectional-tendency radiation. This is directly linked to the perceptual quality descriptor “directional tone colour” ().

References: , .

C. Axial symmetry index

$$S_n(\omega) = \frac{|a_n^0(\omega)|^2}{E_n(\omega)}, \quad (9)$$

which measures the fraction of order- n energy carried by the axially symmetric ($m = 0$) component.

What it shows: $S_n \rightarrow 1$ implies that the radiation at order n is symmetric about the vertical axis; $S_n \rightarrow 0$ implies predominantly non-axisymmetric patterns. For the violin, the broken symmetry introduced by the bass bar and sound post is expected to yield $S_n \ll 1$ for even n .

Representation: Grouped bar charts of S_n per order, per octave band, for the six violins.

References: , .

D. Phase of SH coefficients and inter-modal coupling

What it shows: The phase $\angle a_n^m(\omega)$ encodes the *coupling* between structural modes and the radiation channel (n, m) . When two structural modes overlap in frequency, the relative phase of their contributions to the same SH channel governs constructive vs. destructive interference in specific angular directions. Slow, monotonic phase evolution (close to minimum-phase) indicates lightly coupled, well-separated modes; rapid phase excursions indicate strong modal overlap or internal feedback paths.

Representation: (a) 2D colour maps of unwrapped $\angle a_n^m(\omega)(f)$ in the $(n \cdot (2n + 1) + m + 1, \log f)$ plane; (b) Unit-circle polar plots of $a_n^m(\omega)$ at frequency slices through modal resonances — these display simultaneously the amplitude modulation and phase rotation as frequency sweeps through a resonance.

Comparative value: Phase maps that differ strongly between two violins at the same frequency indicate different modal coupling mechanisms, even if amplitude spectra appear similar. This is therefore a more discriminating fingerprint of the instrument.

References: , , .

E. Phase velocity of SH modes

$$v_\phi^{(n)}(f) = -\frac{2\pi f}{\partial_f \angle a_n^0(\omega)(f)}, \quad (10)$$

computed for the dominant ($m = 0$) component of each order.

What it shows: $v_\phi^{(n)}$ characterises the *dispersive* behaviour of the radiation channel. Departures from the

non-dispersive (free-field) reference $v_\phi = c$ indicate resonant build-up or delay associated with structural resonances, analogous to dispersive wave propagation. The frequency at which $v_\phi^{(n)} \rightarrow \infty$ (phase plateau) marks a resonance of the coupled structural-acoustic system.

Representation: $v_\phi^{(n)}(f)/c$ curves per order, per violin — one panel per order $n = 0, \dots, 4$.

References: , .

F. Radiated power and relative radiation efficiency

$$\eta_{rel,i}(f) = \frac{W_i(f)}{W_{ref}(f)}, \quad W_{ref}(f) = \frac{1}{6} \sum_{i=1}^6 W_i(f), \quad (11)$$

where W_{ref} is the mean power of the ensemble.

What it shows: η_{rel} quantifies how much more or less efficiently instrument i converts bridge force into radiated sound energy at each frequency. Integrated over musical sub-bands (e.g., by string), it provides a single-number loudness proxy.

Representation: Superimposed power spectra $W_i(f)$ (dB re $1 \text{ Pa}^2 \text{ N}^{-2}$) in one-third-octave bands; $\eta_{rel,i}(f)$ as a ratio plot with unity reference.

Comparative value: A violin with $\eta_{rel} > 1$ across a broad band radiates significantly more loudly than the average, which is a primary quality criterion for performers.

References: , , .

G. Modal parameters at structural resonances

What it shows: Frequencies, amplitudes and half-power bandwidths of the main resonances visible on $W(f)$: A0 (air mode), B1[−], B1⁺ (main wood modes), and higher-frequency modes up to 4 kHz. These are the “signature modes” routinely targeted by luthiers.

Representation: Table of modal frequencies, Q -factors and power level at resonance, one row per violin; scatter plots (B1[−] vs. B1⁺ frequency, for example) to visualise clustering.

References: , , .

VI. COMPLEX FAR-FIELD DIRECTIVITY: FREQUENCY DOMAIN

A. Amplitude directivity balloons

What it shows: The amplitude $|D(\theta, \varphi, \omega)|$ at selected frequencies (A0, B1[−], B1⁺, ...) visualises how the sound energy is spatially distributed. At low frequencies the balloon approaches a sphere (monopole); at higher frequencies it develops pronounced lobes and nulls.

Representation: 3D balloon plots rendered on the unit sphere with a colour scale in dB; equatorial and meridional 2D cuts (polar plots) for quantitative reading. Common colour scale across the six violins enables direct comparison.

References: , , .

B. Phase directivity maps

What it shows: The phase $\angle D(\theta, \varphi, \omega)$ at selected frequencies reveals *zones of phase opposition* — directions in which the source radiates in anti-phase relative to a reference direction. Phase inversions in the directivity pattern correspond to nodal lines of the equivalent vibrating surface; they are the angular signature of dipole or higher-order radiation. Two violins with identical amplitude balloons but different phase maps have fundamentally different radiation characters.

Representation: Aitoff equal-area projections of $\angle D(\theta, \varphi, f_0)$ (colour: $-\pi$ to $+\pi$) at modal frequencies; side-by-side comparison for the six violins.

References: , (Ch. 5).

C. Directivity index

$$DI(f) = 10 \log_{10} \left(\frac{4\pi |D_{max}(f)|^2}{W(f)} \right) \quad [\text{dB}], \quad (12)$$

where $D_{max}(f) = \max_{\theta, \varphi} |D(\theta, \varphi, \omega)|$.

What it shows: The DI is a scalar measure of how concentrated the radiated energy is. A high DI indicates that the instrument “beams” its sound in a preferred direction at that frequency, which has direct implications for room acoustics and audience perception.

Representation: $DI(f)$ curves for the six violins, one-third-octave averages, with standard deviation bands.

References: , .

D. Active directivity vector

$$\mathbf{I}(f) = \oint_{S^2} \hat{r} |D(\theta, \varphi, \omega)|^2 d\Omega, \quad (13)$$

a vector pointing in the direction of maximum net energy flux, analogous to the acoustic intensity vector.

What it shows: The direction of $\mathbf{I}$ indicates the *preferred emission axis* of the violin at frequency f . Its magnitude $\|\mathbf{I}\|$ complements the DI: a large $\|\mathbf{I}\|$ with DI close to unity means the emission is concentrated along a single axis. The evolution of the direction of $\mathbf{I}(f)$ with frequency can reveal systematic tilts of the radiation axis linked to the geometry of the internal structure.

Representation: 3D polar plot of the locus of the unit vector $\hat{\mathbf{I}}(f)$ as frequency varies; $\|\mathbf{I}(f)\|/W(f)$ vs. frequency.

References: , .

E. Directional quality factor

$$Q_{dir}(\theta, \varphi) = \frac{f_0}{\Delta f_{3\text{dB}}} \bigg|_{\theta, \varphi}, \quad (14)$$

where f_0 and $\Delta f_{3\text{dB}}$ are the centre frequency and 3 dB bandwidth of each resonance peak in the directional spectrum $D(\theta, \varphi, f)$ at fixed (θ, φ) .

What it shows: Q_{dir} maps the *modal selectivity* in angle: some directions exhibit sharply peaked resonances (high Q , strong coupling between body mode and radiation in that direction), while others show broad plateaus (multiple modes contributing). A violin with very high Q_{dir} in the audience direction will sound very different at frequencies close to resonance vs. between resonances.

Representation: Spherical balloon of $Q_{dir}(\theta, \varphi)$ computed at each identified resonance frequency.

References: , .

F. Angular phase coherence

$$C(\theta_1, \theta_2, f) = \frac{|D(\theta_1, f)\overline{D(\theta_2, f)}|}{|D(\theta_1, f)||D(\theta_2, f)|}, \quad (15)$$

where (θ, φ) indices are abbreviated to θ for clarity.

What it shows: $C \rightarrow 1$ means the two directions are in phase (coherent radiation); $C \rightarrow 0$ means their relative phase is random across the probed bandwidth. Low coherence between, say, the audience direction and the side direction over a broad frequency band implies that the two zones experience acoustically independent source behaviour — they are, in effect, driven by different radiating mechanisms.

Representation: 2D symmetric matrices of $C(\theta_a, \theta_b, f_0)$ for a grid of representative directions and selected modal frequencies.

References: , .

G. Directional group delay

$$\tau_g(\theta, \varphi, f) = -\frac{\partial}{\partial \omega} \angle D(\theta, \varphi, \omega), \quad (16)$$

computed via finite differences after phase unwrapping.

What it shows: $\tau_g > 0$ indicates that the radiation in direction (θ, φ) at frequency f is delayed relative to a non-dispersive source. Peaks of τ_g near structural resonances identify directions in which the body “stores” energy before re-radiating, an indication of high modal Q in that direction. Anomalously large τ_g in a specific direction may signal a defect (e.g., a resonance trapped by a specific structural path).

Representation: Balloon of $\tau_g(\theta, \varphi)$ at modal frequencies; τ_g spectra at a few fixed directions for all six violins.

References: , .

VII. TIME-FREQUENCY ANALYSIS OF DIRECTIVITY

A. Directional spectrograms

What it shows: The short-time Fourier transform (STFT) or continuous wavelet transform (CWT) of the directional signal $D(\theta_0, \varphi_0, f)$ at fixed representative directions (forward, backward, upward, downward, audience, performer) maps the time–frequency energy content of the radiation in those directions. It reveals, for

example, whether radiation in the audience direction is dominated by a brief onset (transient modes) or a sustained tail (slow-decaying resonances).

Representation: Colour spectrograms (time vs. frequency, amplitude in dB) for 4–6 directions, one panel per violin, with reference direction overlay.

References: , .

B. Inter-directional coherence in time-frequency domain

What it shows: Time–frequency coherence between pairs of directions confirms whether modes identified in the frequency domain are simultaneously active in all directions (global body modes) or only in some (localised plate modes).

Representation: Time–frequency coherence maps for selected direction pairs; bar charts of mean coherence per modal frequency band.

References: .

VIII. IMPULSE DIRECTIVITY IN THE FAR FIELD

A. Directional impulse response maps

What it shows: $d(\theta, \varphi, t)$ is the time-domain counterpart of the complex directivity: it describes how the impulsive energy radiated by the violin evolves in angle over time. Early arrivals ($t < 10$ ms) correspond to direct radiation from the excitation point; later arrivals encode modal ring-down and internal reflections. The spatial distribution of the early energy characterises the “attack” character of the instrument; the late-energy distribution characterises the “sustain”.

Representation: 2D maps of $d(\theta_0, \varphi, t)$ as a function of azimuth φ and time t at fixed elevation θ_0 (equatorial and meridional slices); animated 3D balloon over time (supplementary material).

References: , .

B. Hilbert envelope and directional energy

$$e(\theta, \varphi, t) = |\mathcal{H}\{d(\theta, \varphi, t)\}|^2, \quad (17)$$

where $\mathcal{H}$ denotes the Hilbert transform (analytic signal envelope).

What it shows: $e(\theta, \varphi, t)$ provides a smooth, instantaneous measure of the directional energy density, free from rapid oscillations, enabling the identification of direction-dependent decay envelopes.

Representation: Same 2D slice maps as above, now for $10 \log_{10} e$; cumulative energy maps $E_{cum}(\theta, \varphi, t) = \int_0^t e d\tau$.

References: , .

C. Directional centroid time

$$\bar{t}(\theta, \varphi) = \frac{\int_0^\infty t e(\theta, \varphi, t) dt}{\int_0^\infty e(\theta, \varphi, t) dt}. \quad (18)$$

What it shows: $\bar{t}$ is the “centre of gravity” of the radiated energy in each direction. A direction with a large $\bar{t}$ receives energy late, suggesting that the radiation path to that angle is dominated by slow, high- Q resonances. Differences in $\bar{t}(\theta, \varphi)$ between violins directly map to differences in how “bright” or “sustained” the sound feels in each direction.

Representation: Balloon of $\bar{t}(\theta, \varphi)$ for each violin; difference balloon $\bar{t}_i - \bar{t}_{ref}$.

References: , .

D. Directional early-decay time

The early-decay time $EDT(\theta, \varphi)$ is defined as 6 times the time for the energy to drop by 10 dB from its peak in direction (θ, φ) .

What it shows: EDT is a perceptually relevant index: it governs the perceived decay rate of a note. Directional variations of EDT reveal whether the instrument decays uniformly in all directions (an isotropic “room-filling” character) or preferentially in some (directional sustain).

Representation: Balloon of $EDT(\theta, \varphi)$ per violin; boxplots of EDT distributions for the six violins at equal frequencies.

References: , .

E. Directional anisotropy index

$$\sigma_\Omega(t) = \text{std}_{\theta, \varphi} \left[\sqrt{e(\theta, \varphi, t)} \right], \quad (19)$$

the standard deviation of the directional pressure envelope over the sphere at each instant.

What it shows: $\sigma_\Omega(t)$ is a time-resolved measure of how “directional” the sound field is. A high early σ_Ω and a low late σ_Ω indicates a violin that starts with a focused attack and decays into a more diffuse radiation — a pattern associated with clarity of articulation.

Representation: $\sigma_\Omega(t)$ curves for the six violins on the same axes.

References: .

F. Early vs. late energy separation

What it shows: Separating the cumulative directional energy into an early part ($t < t_1$, typically 50 ms, corresponding to modal transients) and a late part ($t > t_1$, corresponding to structural ring-down) isolates two physically distinct radiation regimes. The early directivity is governed by the initial mode shapes excited; the late directivity by the slowest-decaying modes (usually low-order SH components).

Representation: Two balloons (early and late) side by side for each violin; ratio map $E_{early}/E_{late}(\theta, \varphi)$.

References: , , .

IX. MULTI-VIOLIN COMPARISON

A. Complex spherical distance between directivities

$$d_{ij}(\omega) = \|D_i(\cdot, \omega) - D_j(\cdot, \omega)\|_{L^2(S^2)} = \left(\sum_{n,m} |a_n^i(\omega)nm - a_n^j(\omega)nm|^2 \right)^{1/2} \quad (20)$$

and its decomposition into amplitude and phase contributions:

$$d_{ij}^{amp}(\omega) = \| |D_i(\cdot, \omega)| - |D_j(\cdot, \omega)| \|_{L^2}, \quad (21)$$

$$d_{ij}^{ph}(\omega) = (d_{ij}^2 - (d_{ij}^{amp})^2)^{1/2}. \quad (22)$$

What it shows: The full complex distance d_{ij} captures both amplitude and phase differences. The decomposition (21)–(22) allows one to determine whether two violins differ primarily in *how much* they radiate in a given direction (amplitude) or in the *timing/phase* of that radiation. Phase-dominant differences ($d_{ij}^{ph} \gg d_{ij}^{amp}$) at a given frequency suggest different modal coupling mechanisms despite similar radiated power.

Representation: (a) 6×6 symmetric matrices of $d_{ij}(f)$ for selected frequencies and octave bands; (b) Separate matrices for d_{ij}^{amp} and d_{ij}^{ph} ; (c) Dendrograms derived from the mean distance across the studied bandwidth.

References: , .

B. Spherical cross-correlation

$$\rho_{ij}(\omega) = \frac{\langle D_i(\cdot, \omega), D_j(\cdot, \omega) \rangle_{L^2}}{\|D_i\| \|D_j\|}, \quad (23)$$

where the inner product is Hermitian.

What it shows: ρ_{ij} is a complex number: its modulus is the amplitude similarity (1 = identical shapes) and its argument is the mean phase offset between the two directivity patterns. A pair with $|\rho_{ij}| \approx 1$ and $\angle \rho_{ij} \approx 0$ are acoustically near-twins; a small modulus with a non-trivial phase implies structurally different radiation patterns.

Representation: Polar scatter plots of ρ_{ij} in the complex plane, per frequency band; heat maps of $|\rho_{ij}|$ and $\angle \rho_{ij}$ separately.

References: (Ch. 5), .

C. Complex PCA on SH coefficient vectors

The data matrix $\mathbf{A}(f) \in \mathbb{C}^{6 \times 64}$ stacks the SH coefficient vectors of the six violins at frequency f . A complex principal component analysis (PCA) or singular value decomposition (SVD) of $\mathbf{A}$ yields the dominant directions of inter-violin variability.

What it shows: The first few principal components identify the “axes of acoustic variation”: e.g., one component may separate violins by radiated power, another by the orientation of the dipole lobe, another by the modal

coupling phase. Projecting each violin onto the first two PCs provides a low-dimensional “acoustic fingerprint” map where proximity implies acoustic similarity.

Representation: 2D scatter plot of the six violins in the (PC1, PC2) plane at selected frequencies; variance explained by each PC as a bar chart; loadings of PC1 and PC2 displayed as balloons on the sphere.

References: , .

D. Hierarchical clustering

Agglomerative hierarchical clustering (Ward linkage) is applied to the mean d_{ij} matrix across the full bandwidth, and separately in sub-bands (low: 80–500 Hz; mid: 500–2000 Hz; high: 2–10 kHz).

What it shows: The dendrogram reveals whether violins cluster according to the lutherie variable of interest. Band-limited clustering shows whether groupings change with frequency, which would indicate that certain lutherie choices affect only specific frequency ranges of the radiation.

Representation: Dendrograms for the full band and for each sub-band; consistency matrix (whether pair (i, j) clusters together across all bands).

References: , .

X. INTERPRETATION: RADIATION SIGNATURES AND LUTHERIE TECHNIQUE

[This section interprets the comparative results of Sections V–IX in the light of the specific lutherie variables that distinguish the six violins. Key questions include: (i) Which lutherie variables produce significant differences in radiated power $W(f)$ and at which frequencies? (ii) Do wood graduation choices affect the modal order distribution $E_n(f)$ and therefore the directional complexity? (iii) Does varnish type correlate with differences in modal Q , as reflected in τ_g and EDT maps? (iv) Are differences primarily of amplitude or phase character, and what does this imply for the room acoustic impression? (v) Do acoustically “better” violins (by expert judgement) cluster together in the PCA space?

These questions will be addressed in detail once experimental results are available. Placeholder figures: Fig. X – Balloon comparison at B1⁺; Fig. Y – PCA scatter across frequency bands; Fig. Z – Phase distance matrix heatmap.]

XI. CONCLUSION

This paper has presented a comprehensive framework for the measurement, decomposition and comparison of the three-dimensional complex radiation directivity of six violins built with identical external geometry but different lutherie techniques. The HELS-based spherical-harmonic expansion up to order 7 provides a compact, physically interpretable representation of the radiation field over the full musical frequency range. A hierarchy of scalar and vectorial indices – from radiated power

and modal energy spectra to active directivity vector, angular coherence, group delay, impulse centroid time and complex inter-violin distance – enables a multi-scale comparative analysis that takes full account of both the amplitude and the phase of the radiation.

Key findings [to be completed]: . . .

The methodology developed here is general and applicable to any musical instrument or vibrating structure for which impact hammer excitation is representative. Future work will extend the analysis to bowed-string excitation and to a larger ensemble of instruments.

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